

13. Implement the Car Price Prediction Model using Python

Aim

To implement a **Car Price Prediction model** using an appropriate machine learning algorithm.

Software Required

- Python 3.x
- Pandas
- Scikit-learn

Procedure

1. Create a car dataset containing age, mileage and engine size.
2. Separate features and target price.
3. Divide the dataset into training and testing data.
4. Apply Linear Regression.
5. Train the model.
6. Predict the car price.
7. Calculate the prediction error.

Code

```
import pandas as pd

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error

data = {
    'Age': [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
    'Mileage': [10, 20, 30, 40, 50, 60, 70, 80, 90, 100],
    'Engine': [2.0, 2.0, 1.8, 1.8, 1.6,
               1.6, 1.5, 1.5, 1.4, 1.4],
    'Price': [20, 18, 16, 15, 13, 11, 10, 8, 7, 5]
}

df = pd.DataFrame(data)
```

```

X = df[['Age', 'Mileage', 'Engine']]
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Predicted Prices:")
print(y_pred)
print("\nMean Squared Error:",
      mean_squared_error(y_test, y_pred))
sample = [[3, 30, 1.8]]
print("\nPredicted price for new car:",
      model.predict(sample)[0], "Lakhs")

```

Output:

```

Predicted Prices:
[ 6.41990291 18.33252427]

```

```

Mean Squared Error: 0.2235425110755039

```

```

Predicted price for new car: 16.342233009708742 Lakhs
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning:
  warnings.warn(

```

Result

Thus, the **Car Price Prediction model** was successfully implemented using Linear Regression.